

120 Days of Quant Finance

Week 3 – Linear Algebra

Day 1: Vectors, Matrices, and Portfolio Weights

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Introduction

Linear algebra provides the mathematical language for representing and manipulating high-dimensional objects. In quantitative finance, portfolios, asset returns, and risk measures are naturally expressed using vectors and matrices.

This section introduces vectors and matrices from a mathematical perspective and establishes the foundational representations used in portfolio modeling.

Vectors

A **vector** is an ordered collection of real numbers. Formally, a vector $\mathbf{x}$ in $\mathbb{R}^n$ is written as:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The dimension n specifies the number of components and determines the space in which the vector resides.

Vector Operations

Given vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and scalar $\alpha \in \mathbb{R}$:

- **Addition:**

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

- **Scalar multiplication:**

$$\alpha \mathbf{x} = \begin{pmatrix} \alpha x_1 \\ \vdots \\ \alpha x_n \end{pmatrix}.$$

These operations preserve dimensionality and form the basis of linear combinations.

Dot Product

The **dot product** (or inner product) of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as:

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

The dot product produces a scalar and satisfies symmetry and linearity. It plays a central role in measuring projections, similarity, and aggregation of components.

Vector Norm

The **Euclidean norm** of a vector $\mathbf{x}$ is defined as:

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

The norm measures the magnitude of a vector and corresponds to its distance from the origin in $\mathbb{R}^n$.

Matrices

A **matrix** is a rectangular array of real numbers. An $n \times m$ matrix $\mathbf{A}$ is written as:

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{pmatrix}.$$

Matrices represent linear transformations that map vectors from one space to another.

Matrix–Vector Multiplication

Given $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{x} \in \mathbb{R}^m$, the product $\mathbf{y} = \mathbf{Ax}$ yields a vector $\mathbf{y} \in \mathbb{R}^n$.

Matrix–vector multiplication is defined component-wise as:

$$y_i = \sum_{j=1}^m a_{ij}x_j.$$

Dimension compatibility is a necessary condition for multiplication to be defined.

Special Matrices

Identity Matrix

The identity matrix $\mathbf{I}_n$ is defined as:

$$\mathbf{I}_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

It satisfies $\mathbf{I}_n \mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.

Diagonal Matrix

A diagonal matrix has nonzero elements only along its main diagonal:

$$\mathbf{D} = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

Diagonal matrices are used to scale vector components independently.

Portfolio Weights as a Vector

A portfolio of n assets is represented by a **weight vector**:

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix},$$

where w_i denotes the fraction of total capital allocated to asset i .

Budget Constraint

The fully invested portfolio constraint is expressed as:

$$\sum_{i=1}^n w_i = 1 \quad \Longleftrightarrow \quad \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{1}$ denotes the vector of ones.

Portfolio Return

Let $\mathbf{r}$ denote the vector of asset returns. The portfolio return is given by the dot product:

$$R_p = \mathbf{w}^\top \mathbf{r}.$$

This formulation highlights the linear relationship between asset returns and portfolio weights.

Portfolio Variance

Let Σ denote the covariance matrix of asset returns. The portfolio variance is expressed as the quadratic form:

$$\text{Var}(R_p) = \mathbf{w}^\top \Sigma \mathbf{w}.$$

Quadratic forms play a central role in risk measurement and portfolio optimization.

Conclusion

Vectors and matrices provide a compact and expressive framework for modeling portfolios and their associated risk and return. Representing portfolios as vectors and dependencies as matrices enables systematic analysis and lays the foundation for modern portfolio theory and optimization techniques.